

Feature Specification Document

Version: 1.0

Document Control

Version	Date	Author	Description
1.0	2026-05-12	Hardik Patel	Initial Version

Reviewers

Name	Role	Approval Date
Hardik Patel	CTO	2026-05-13

Approvers

Name	Role	Approval Date
Hardik Patel	CTO	2026-05-14

Feature Overview

Provide a high-level overview of the feature.

Include:

- Feature purpose
- User value
- Business impact

Example:

The Checkout feature allows users to review cart items, select delivery address, complete payment, and place orders successfully.

Business Context

Explain why this feature is important.

Example:

A simplified checkout experience is expected to reduce cart abandonment and improve conversion rates.

User Roles

Define users interacting with this feature.

User Role	Description
Customer	Places orders
Admin	Monitors order processing
Support Agent	Assists with failed orders

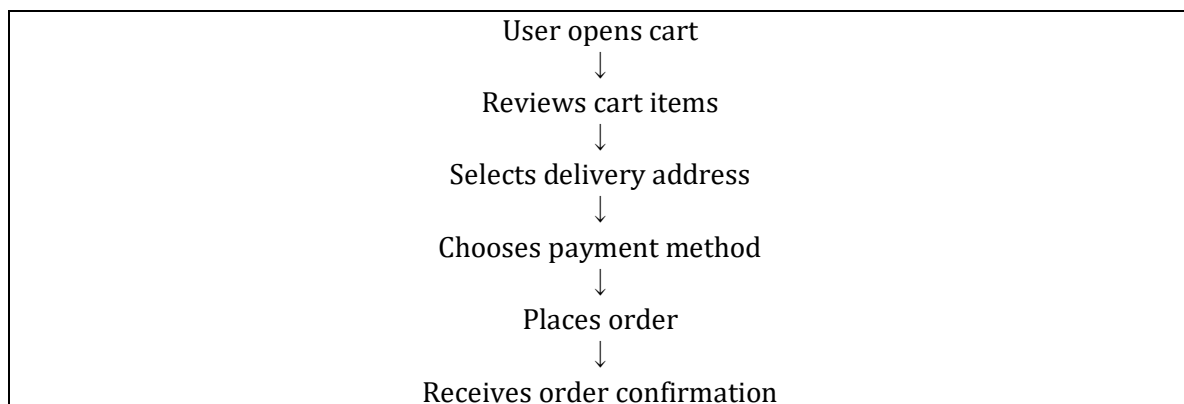
Feature Goals

Goal ID	Goal
FG-001	Reduce checkout completion time
FG-002	Improve payment success rate
FG-003	Improve order placement reliability

User Flow

Describe the end-to-end workflow.

Example Flow:



Functional Requirements

Define feature behavior in detail.

Requirement ID	Description
FR-001	Users should be able to review cart items
FR-002	Users should be able to modify cart quantity
FR-003	Users should be able to select saved addresses
FR-004	Users should be able to place orders securely
FR-005	System should generate order confirmation

Non-Functional Requirements

Define quality expectations.

Requirement ID	Description
NFR-001	Checkout page should load under 2 seconds
NFR-002	Payment APIs should maintain 99.9% uptime
NFR-003	Sensitive payment data must be encrypted
NFR-004	System should support high traffic during sales events

UI/UX Expectations

Describe important user experience considerations.

Example:

- Checkout steps should be minimal
- Error messages should be user-friendly
- Mobile responsiveness is required
- Payment failure handling should be clear

Validation Rules

Define input validation behavior.

Field	Validation Rule
Email Address	Must be valid email format
Phone Number	Must contain valid country code
Payment Method	Required before order placement
Delivery Address	Mandatory for checkout

Error Handling

Define expected error scenarios.

Scenario	Expected Behavior
Payment failure	Display retry option
Inventory unavailable	Inform user immediately
Session expired	Redirect to login
Invalid address	Show validation error

Edge Cases

Document uncommon but important scenarios.

Edge Case	Expected Handling
Item goes out of stock during checkout	Prevent order placement
Duplicate payment request	Prevent duplicate order creation
Network interruption	Allow retry without losing cart

API Considerations

List APIs related to the feature.

API	Purpose
POST /checkout	Create checkout session
POST /orders	Place order
GET /inventory	Validate stock availability
POST /payments	Process payment

Database Considerations

Define impacted entities and data changes.

Entity	Purpose
Cart	Stores selected products
Orders	Stores order details
Payment	Stores transaction details
Address	Stores delivery information

Security Considerations

Define security expectations.

Area	Requirement
Authentication	Users must be authenticated
Authorization	Users can access only their own carts
Encryption	Payment data must be encrypted
Audit Logging	Order activities should be logged

Dependencies

Dependency	Description
Payment Gateway	Required for payment processing
Inventory Service	Required for stock validation
Notification Service	Required for order confirmation

Success Metrics

Metric	Target
Checkout Success Rate	98%

Cart Abandonment Rate	Reduce by 20%
Payment Failure Rate	Under 1%

Risks

Risk ID	Description	Impact
RISK-001	Payment gateway downtime	High
RISK-002	High traffic during sales events	Medium
RISK-003	Inventory synchronization delays	Medium

Open Questions

Question ID	Question	Owner
Q-001	Should guest checkout be supported?	Product Team
Q-002	Should multiple payment methods be allowed?	Business Team

Related Documents

Document	Purpose
BRD	Defines business requirements
PRD	Defines product requirements
API Specification	API contract definitions
DB Schema	Data model definitions
Architecture Documentation	System architecture and infrastructure